

Sima Ghafoori

Ph.D. Candidate, Electrical Engineering

Translational Neurorobotics Lab • University of Rhode Island • Kingston, RI, USA
Sima.ghafoori@uri.edu • +1 (401) 212-0748 • [LinkedIn](#) • [GitHub](#) • [Google Scholar](#)

RESEARCH STATEMENT

Ph.D. researcher in Electrical Engineering specializing in real-time robotic systems, adaptive control, state estimation, and learning-based methods for human-robot interaction. Research integrates biosignals (EEG/EMG), force sensing, and embedded hardware into closed-loop robotic platforms for upper-limb motor rehabilitation. Broader interests span assistive robotics, brain-computer interfaces, and the translation of neuroscience-informed signal processing into usable rehabilitation technologies.

EDUCATION

Ph.D. in Electrical Engineering

University of Rhode Island

Kingston, RI, USA

Jan. 2023 – 2027 (expected)

Dissertation: *Design and Control of Biosignal-Based Planar Robotic Platforms for Upper-Limb Motor Rehabilitation.*

Advisor: Dr. Reza Abiri, Translational Neurorobotics Lab, Department of Electrical, Computer, and Biomedical Engineering. GPA: 3.84 / 4.0

M.S. in Electrical Engineering

Shahid Beheshti University of Medical Sciences

Tehran, Iran

Sep. 2019 – Dec. 2021

Thesis: *Predicting Conversion from Mild Cognitive Impairment to Alzheimer's Disease by Integrating Functional MRI and Clinical Data Using Deep Learning Methods.*

GPA: 4.0 / 4.0

B.Sc. in Electrical Engineering

University of Isfahan

Isfahan, Iran

Sep. 2015 – Sep. 2019

Senior Project: *Design of an Online Computer-Aided Diagnostic System for Seizure Prediction Using an Optimal Number of Electrodes.*

GPA: 3.64 / 4.0

RESEARCH EXPERIENCE

Graduate Research Assistant (Ph.D.)

Translational Neurorobotics Lab, University of Rhode Island

Kingston, RI, USA

Jan. 2023 – Present

- Enhanced the stability and precision of a magnetically actuated upper-limb rehabilitation platform by implementing a closed-loop Extended Kalman Filter (EKF) for real-time, sensorless state estimation and control.
- Developed a Python interface for real-time visualization and trajectory control of the platform via Arduino-based feedback.

- Advanced the decoding of motor intentions by constructing brain-signal feature-extraction pipelines based on bispectrum, cross-bispectrum, and phase-coupling metrics.
- Designed a real-time closed-loop robotics software framework integrating EEG/EMG acquisition, force/torque sensing, and a 4-DOF linkage robot for end-to-end data collection and control.
- Implemented a biosignal-driven assistive control algorithm that decodes human intent from EEG/EMG to command the robot in real time.
- Engineered a physics-based PyBullet simulation of a 4-DOF rehabilitation robot and interfaced it with a custom load-cell joystick, enabling interactive force-controlled human-in-the-loop testing.
- Developed an adaptive admittance control framework for the 4-DOF rehabilitation robot.
- Contributed to signal-processing pipelines integrating wavelet, ICA, and feature extraction for comparative analysis of EEG and tEEG signal performance in movement decoding.
- Collaborated on a self-training Deep Q-Network (DQN) framework that enhances biosignal-driven robotic control accuracy under noisy input conditions.

Graduate Research Intern

Tehran, Iran

School of Electrical and Computer Engineering, University of Tehran

2021 – 2022

- Explored advanced biosignal-processing techniques and their integration with ML/DL algorithms to enhance robotic control and human-robot interaction.

Graduate Research Assistant (M.S.)

Tehran, Iran

Department of Biomedical Engineering and Medical Physics, SBMU

2019 – 2021

- Advanced early Alzheimer's prediction research by processing and analyzing fMRI/MRI data with deep learning architectures (CNN, LSTM, CNN-LSTM) to detect mild cognitive impairment (MCI).
- Developed Arduino-based systems integrating ECG signals, Bluetooth, and ESP8266 modules for biosignal acquisition, cloud transmission, and classification using MATLAB and ML models.

SELECTED PUBLICATIONS ([Google Scholar](#))

Journal Articles

- [J1] **S. Ghafoori**, A. Rabiee, M. Jouaneh, and R. Abiri, "*A Novel Seamless Magnetic-Based Actuating Mechanism for End-Effector-Based Robotic Rehabilitation Platforms*," IEEE Robotics and Automation Letters, vol. 10, pp. 4516–4523, 2025. **[Published]**
- [J2] A. Rabiee, **S. Ghafoori**, A. Cetera, W. Besio, and R. Abiri, "*A Comparative Study of Conventional and Tripolar EEG for High-Performance Reach-to-Grasp BCI Systems*," IEEE Transactions on Biomedical Engineering, 2025. **[Published]**
- [J3] **S. Ghafoori**, A. Cetera, A. Rabiee, M. H. Farhadi, R. Singh, M. Furmanek, Y. Shahriari, and R. Abiri, "*Cross-Frequency Bispectral EEG Analysis of Reach-to-Grasp Planning and Execution*," Computers in Biology and Medicine, 2026. **[Published]**
- [J5] A. Cetera, **S. Ghafoori**, A. Rabiee, Y. Shahriari, R. Abiri, "*Macroscopic EEG Reveals Discriminative Low-Frequency Oscillations in Plan-to-Grasp Visuomotor Tasks*," Journal of Neural Engineering, 2026. **[Revision]**
- [J6] **S. Ghafoori**, et al., "*Geometry-Aware Adaptive Admittance Control for Upper-Limb Rehabilitation Robotics*," Manuscript in preparation, 2026. **[In Preparation]**

Conference Proceedings

- [C1] **S. Ghafoori**, A. Rabiee, A. Cetera, and R. Abiri, "*Bispectrum Analysis of Noninvasive EEG Signals Discriminates Complex and Natural Grasp Types*," IEEE Engineering in Medicine and Biology Society (EMBC), 2024. **[Published]**
- [C2] A. Rabiee, **S. Ghafoori**, A. Cetera, and R. Abiri, "*Wavelet Analysis of Noninvasive EEG Signals Discriminates Complex and Natural Grasp Types*," IEEE Engineering in Medicine and Biology Society (EMBC), 2024. **[Published]**
- [C3] A. Cetera, A. Rabiee, **S. Ghafoori**, and R. Abiri, "*Classification of Emerging Neural Activity from Planning to Grasp Execution using a Novel EEG-Based BCI Platform*", IEEE Engineering in Medicine and Biology Society (EMBC), 2024. **[Published]**
- [C4] A. Rabiee, **S. Ghafoori**, X. Bai, S. Ostadabbas, and R. Abiri, "*Streams: An Assistive Multimodal AI Framework for Empowering Biosignal-Based Robotic Controls*," Artificial Intelligence, Robotics and Control (AIRC), 2025. **[Published]**

Book Chapters

- [B1] M. H. Farhadi, A. Rabiee, **S. Ghafoori**, A. Cetera, W. Xu, and R. Abiri, "*Human-Centered Shared Autonomy for Motor Planning, Learning, and Control Applications*," in *Bridging the Gap between Mind and Machine: Exploring the Future of Human-AI-Neurotechnology Integration*, Springer Nature Switzerland, 2025, pp. 259–297. **[Published]**

Supported by grants

- US National Science Foundation (NSF) 2023. Award ID: 2245558.
- National Institute of General Medical Sciences of the National Institutes of Health (NIH). Grant number P20GM103430

CONFERENCE PRESENTATIONS & TALKS

- [P1] **S. Ghafoori**, A. Rabiee, M. Jouaneh, and R. Abiri. "*Development of a novel seamless magnetic-based actuated planar robotic platform for upper limb extremity rehabilitation*," Poster presentation, Society for Neuroscience (SFN), Washington, D.C., USA, 2023.
- [P2] **S. Ghafoori**, A. Rabiee, A. Cetera, and R. Abiri. "*Bispectrum Analysis of Noninvasive EEG Signals Discriminates Complex and Natural Grasp Types*," poster presentation, 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Orlando, FL, USA, 2024.
- [P3] **S. Ghafoori**. "*Getting Arms Back in Action: A Rehab Robot Story*," 3-minute lightning talk, Smart Health & Wearables Symposium 2024, University of Rhode Island, Kingston, RI, USA, 2024.
- [P4] **S. Ghafoori**, A. Rabiee, A. Cetera, Y. Shahriari, and R. Abiri. "*Bispectrum Analysis of Noninvasive EEG Signals Discriminates Complex and Natural Grasp Types*," poster presentation, Department of ECBE Graduate Student Poster Competition, University of Rhode Island, Kingston, RI, USA, 2024.

MENTORSHIP, LEADERSHIP & AWARDS

- Mentored undergraduate and senior capstone teams in developing a 1-DOF upper-limb rehabilitation platform.
- Supervised the design and implementation of a custom force-sensing joystick for human-in-the-loop testing.

- Guided the development of PyBullet simulations for a 4-DOF rehabilitation robot.
- Organized and led a high school internship program focused on robotics and neurotechnology.
- Contributed to and co-led research and development efforts on an NSF-funded project (*Award No. 2245558*) integrating biosignal processing with a 4-DOF rehabilitation platform.
 - Outstanding Researcher Award - ECBE Graduate Student Poster Competition, University of Rhode Island (2026).
 - Best Poster Award - ECBE Graduate Student Poster Competition, University of Rhode Island (2024).

TECHNICAL SKILLS

Programming	C, C++, Python, MATLAB, JavaScript
AI / ML	TensorFlow, Keras, PyTorch, Scikit-learn, Scikit-Optimize; reinforcement learning; data analysis with Pandas, Matplotlib, Seaborn
Robotics & Control	Adaptive control, admittance control, real-time control, path planning, system identification; 1-DOF belt-driven rehabilitation robot, 4-DOF rehabilitation arm
Robotics Simulation	PyBullet, MATLAB / Simulink, Simscape
Estimation & Algorithms	Extended Kalman Filter (EKF), ERP, ICA, PCA, PSD, wavelet, bispectrum, cross-bispectrum
Signal Processing	Filtering and artifact removal, PSD estimation (Welch), time-frequency analysis (STFT, wavelets), nonlinear feature extraction (bispectrum / phase coupling); MATLAB and Python (NumPy, SciPy, MNE, PyWavelets)
Tools & Environments	Linux (Ubuntu), VS Code, Visual Studio, PyCharm, SolidWorks, Altium, LTSpice, DipTrace, ClearPath, Pro-Motion, HitBotStudio, SPM

LANGUAGES

English (Professional working proficiency) • **Persian** (Native)

REFERENCES

Available upon request.